

Haoxuan Cheng

Curriculum Vitae

School of Mathematical Sciences, Fudan University
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Education

Fudan University — Direct-entry PhD program in mathematics, 2025–present.

South China Normal University — Previous studies in mathematics.

Research interests

Discrete geometric analysis; Ricci curvature on graphs; spectral graph theory; geometric analysis on manifolds, including volume growth, stochastic processes, and curvature flows.

Support

Class B funding, National High-Level Talent Training Center at Fudan University.

Talks and poster presentations

- **Discrete Einstein Metrics on Trees.** Poster presentation, *Geometric and analytic functional inequalities on discrete spaces*, Tsinghua Sanya International Mathematics Forum, Sanya, China. Conference: 14–18 September 2026. [Conference page](#).
- **Monotonicity and Limiting Behavior of Einstein Trees I: Leaf Attachment.** Beijing Institute of Mathematical Sciences and Applications (BIMSA), 26 June 2026. [Talk page](#).
- **A Classification of Positive-Curvature Discrete Einstein Metrics on Trees.** BIMSA, 20 May 2026. [Seminar listing](#).

Research software

Graph Geometry Research Workspace. A workbench for organizing mathematical research with Codex, including proof attempts, dependencies, computational evidence, and project templates. [Public repository](#).

Updated 6 October 2026

Preprints

10. **A counterexample to Yau's bounded-mean-curvature embedding question**
Haoxuan Cheng. 2026. [arXiv:2610.00153](#).
9. **Umbilic slopes and cubic Weingarten surfaces**
Haoxuan Cheng. 2026. [arXiv:2609.19188](#).
8. **Bounded curvature manifolds without Euclidean isometric immersions of bounded mean curvature**
Haoxuan Cheng. 2026. [arXiv:2609.19168](#).
7. **Unbounded normalized scalar curvature integrals in dimension four**
Haoxuan Cheng. 2026. [arXiv:2609.10323](#).
6. **Singular Rotational Self-Similar Tori for Odd σ_k -Curvature Flows**
Haoxuan Cheng, Junqi Lai, Guoxin Wei. 2026. [arXiv:2609.01346](#).
5. **Volume Growth and Recurrence of Fractional Powers of the Laplace--Beltrami Operator**
Haoxuan Cheng. 2026. [arXiv:2608.26509](#).
4. **Edge Subdivision and the Perron Eigenvalue of Tree Ricci Matrices**
Shuliang Bai, Haoxuan Cheng, Bobo Hua. 2026. [arXiv:2605.30949](#).
3. **Spectral Monotonicity under Leaf Attachment and Limiting Behavior in Discrete Einstein Trees**
Shuliang Bai, Haoxuan Cheng, Bobo Hua. 2026. [arXiv:2605.23379](#).
2. **Positive-Curvature Discrete Einstein Metrics on Trees**
Haoxuan Cheng. 2026. [arXiv:2605.20862](#).
1. **Discrete Einstein metrics on trees**
Shuliang Bai, Haoxuan Cheng, Bobo Hua. 2026. [arXiv:2604.22449](#).